

Exp No: 9 Date:	CPM -Analysis
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AIM:

To perform Critical Path Method (CPM) Analysis for project scheduling and management using Python.

PROCEDURE:

Install Libraries: pip install pandas networkx matplotlib

Define the Problem: Identify tasks, their durations, and dependencies.

Set Up in Python: Use NetworkX to represent tasks and dependencies. Apply topological sort to find the critical path.

Run the Code: Execute Python code to calculate the critical path and project duration.

Analyze Output: Interpret results and verify against the project schedule.

PROBLEM -1:

Activity	Duration	Dependencies
A	2	-
B	3	A
C	2	A
D	4	B,C
E	3	D

Code:

```
import
networkx

as nx tasks

= {
    'A': {'duration': 2, 'dependencies': []},
    'B': {'duration': 3, 'dependencies': ['A']},

    'C': {'duration': 2, 'dependencies': ['A']},
    'D': {'duration': 4, 'dependencies': ['B', 'C']},
    'E': {'duration': 3, 'dependencies': ['D']}
}

G = nx.DiGraph()

for task, details in tasks.items(): G.add_node(task,
    duration=details['duration']) for dep in details['dependencies']:
    G.add_edge(dep, task)

# Topological order
order = list(nx.topological_sort(G))

# Earliest Start (ES) and Finish
(EF) ES, EF = {}, {}
for node in order:
    ES[node] = max([EF.get(pred, 0) for pred in G.predecessors(node)], default=0) EF[node]
    = ES[node] + G.nodes[node]['duration']

# Latest Finish (LF) and Start (LS)
LF, LS = {}, {}
max_time = max(EF.values())

for node in reversed(order):
    LF[node] = min([LS.get(succ, max_time) for succ in G.successors(node)],
    default=max_time)
    LS[node] = LF[node] - G.nodes[node]['duration']

# Critical Path
critical_path = [node for node in G.nodes if ES[node] == LS[node]]

print("Earliest start times:", ES) print("Earliest
finish times:", EF) print("Latest start times:",
```

```
LS) print("Latest finish times:", LF)
print("Critical path:", critical_path)
```

Output:

```
Earliest start times: {'A': 0, 'B': 2, 'C': 2, 'D': 5, 'E': 9}
Earliest finish times: {'A': 2, 'B': 5, 'C': 4, 'D': 9, 'E': 12}

Latest start times: {'A': 0, 'B': 2, 'C': 3, 'D': 5, 'E': 9}
Latest finish times: {'A': 2, 'B': 5, 'C': 5, 'D': 9, 'E': 12} Critical path: ['A', 'B', 'D', 'E']
```

RESULT:

Thus, the Critical Path Method (CPM) for scheduling project activities and calculating the critical path using Python was executed and verified Successfully.